

Equivalent Ratios

Lesson 3-4

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



3 : 2

3 red apples : 5 green apples

Ratio

Write the definition:

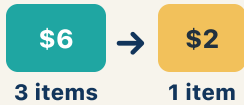
$$2:3 = 4:6$$

×2 → same comparison

2:3 and 4:6 both mean 2 out of every 3, so $\frac{2}{3} = \frac{4}{6}$

Equivalent ratios

Write the definition:



60 miles per 1 hour

Rate

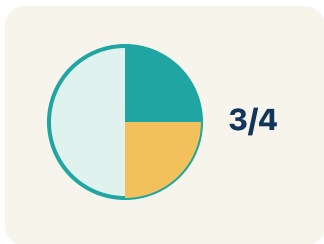
Write the definition:

$$\frac{1}{2} = \frac{2}{4}$$

$$\frac{2}{3} = \frac{4}{6}$$

Proportion

Write the definition:



$12:18 \rightarrow$ divide both by 6 $\rightarrow 2:3$

Simplify

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find and check equivalent ratios using multiplication and division.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Ratio

Equivalent ratios

Rate

Proportion

Simplify

1 A comparison of two quantities is a _____.

2 _____ — Two ratios that mean the same thing.

3 A ratio that compares two quantities with different units, like miles per hour, is a _____.

4 An equation showing that two ratios are equal is a _____.

5 Writing a ratio in its smallest equal form, like $6:9 = 2:3$, is to _____ it.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Which ratio is equivalent to 2:5?

- A. 4:10
- B. 3:6
- C. 2:10
- D. 5:2

Step 1 Multiply both parts by 2: $2 \times 2 = 4$ and $5 \times 2 = 10$, so 4:10 is equivalent to 2:5.

 **Answer:** A. 4:10

 We do — together

Solve this one with your class using the same steps.

Problem: A recipe uses 4 eggs for every 6 cups of flour. Which ratio is equivalent?

- A. 2:3
- B. 8:10
- C. 4:3
- D. 6:4

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Which ratio is NOT equivalent to 6:9?

- A. 3:5
- B. 2:3
- C. 12:18
- D. 18:27

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Chef Montoya's dressing uses 3 spoons of oil to 4 spoons of vinegar (3:4). Which recipe card shows an EQUIVALENT ratio?

- A. 9:12
- B. 4:3
- C. 3:8
- D. 6:7

Show your work:

2. The lemonade station mixes 5 cups of lemon juice to 2 cups of sugar (5:2). Which batch keeps the SAME taste?

- A. 15:6
- B. 10:6
- C. 5:4
- D. 7:4

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A store sells 3 pens for \$2. Another store sells 9 pens for \$6. Are these equivalent ratios? If so, prove it using at least two different methods (simplifying, scaling, or cross-multiplying).

Show your work:

Reflect — Exit Ticket

A store sells 5 notebooks for \$8. At this rate, how much would 15 notebooks cost?

- A. \$24
- B. \$18
- C. \$20
- D. \$40

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Ratio
2. **Notes 2:** Equivalent ratios
3. **Notes 3:** Rate
4. **Notes 4:** Proportion
5. **Notes 5:** Simplify
6. **We Try (notes):** A. $2:3$ — *Divide both by 2: $4 \div 2 = 2$ and $6 \div 2 = 3$. The ratio $2:3$ is equivalent to $4:6$.*
7. **You Try (notes):** A. $3:5$ — $6:9$ simplifies to $2:3$. Check each: $2:3$ ✓, $12:18 = 2:3$ ✓, $18:27 = 2:3$ ✓. But $3:5$ does not simplify to $2:3$.
8. **Try It 1:** A. $9:12$ — *Multiply both parts of $3:4$ by 3: $3 \times 3 = 9$ and $4 \times 3 = 12$, so $9:12$ is equivalent to $3:4$.*
9. **Try It 2:** A. $15:6$ — *Multiply both parts of $5:2$ by 3: $5 \times 3 = 15$ and $2 \times 3 = 6$, so $15:6$ is equivalent to $5:2$.*
10. **Exit Ticket:** A. \$24 — *15 is 3 times 5, so the cost is $3 \times \$8 = \24 . The ratio $5:8$ scales to $15:24$.*